

Tutorial 8: Larmor formula, antennae, multipole radiation

Question 1

- (a) Suppose an electron decelerates from some initial velocity $v_0 \ll c$ to zero at a constant rate a . What fraction of its initial kinetic energy is lost to radiation?
- (b) To put some numbers on it, suppose the initial velocity is thermal $\sim 10^5 \text{ ms}^{-1}$ and the electron travels a distance $30 \text{ \AA}$. What can you conclude about radiation losses for electrons in an ordinary conductor?

[Griffiths 11.13]

Question 2

In the Bohr model of hydrogen, the electron in its ground state travels in a circle of radius $0.5 \text{ \AA}$, held in orbit by the Coulomb attraction of the proton. According to classical electrodynamics, this electron should radiate, and hence spiral in to the nucleus.

- (a) Show that $v \ll c$ for most of the trip (so you can use the Larmor formula).
- (b) Calculate the lifespan of Bohr's atom. (Assume each revolution is essentially circular.)

[Griffiths 11.14]

Question 3

A positive charge q is fired directly at a distant positive charge Q (which is held stationary). It has an initial velocity $v_0 \ll c$. It comes in, decelerates to $v = 0$, and returns out to infinity. Find the fraction of its initial kinetic energy that is radiated away. Assume that you can ignore the effect of radiative losses on the motion of the particle.

[Answer: $\frac{16}{45} \frac{q}{Q} \left(\frac{v_0}{c}\right)^3$]

Question 4

Consider the centre-fed linear dipole with current $I(z, t) = I_0 \sin(kd - k|z|)e^{-i\omega t}\hat{z}$ as discussed in lectures.

- (a) Evaluate the time-averaged angular distribution of power in the limit where the radiation wavelength is much larger than the antenna length, $\lambda \gg d$. This antenna is called “electrically short”.
- (b) Use the continuity equation $\nabla \cdot \mathbf{j} = -\dot{\rho}$ to find the charge density and hence the dipole moment of the antenna. Take the long-wavelength limit and use this to evaluate the time-averaged angular distribution of power radiated by a point electric dipole:

$$\left\langle \frac{dP}{d\Omega} \right\rangle = \frac{\mu_0 p_0^2 \omega^4 \sin^2 \theta}{32\pi^2 c}.$$

Compare with part (a).

[Zangwill 20.10]